Writing translations

Translations are written as vectors. The top number shows the horizontal distance an object moves, while the bottom number shows the vertical distance moved. The two numbers are contained within a set of parentheses. Each translation can be numbered—for example, T_1, T_2, T_3—to make it clear which one is being referred to if more than one translation is shown.

translation number

distance moved horizontally

$$T_1 = \begin{pmatrix} 6 \\ 0 \end{pmatrix}$$

distance moved vertically

△ **Translation T_1**
To move triangle ABC to position $A_1B_1C_1$ each point moves 6 units horizontally, but does not move vertically. The vector is written as above.

translation number

distance moved horizontally

$$T_2 = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$$

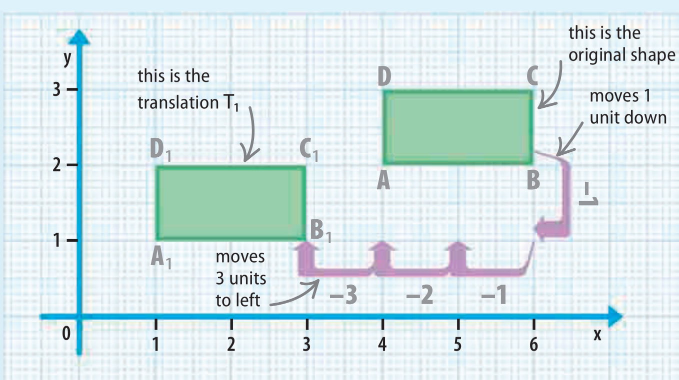
distance moved vertically

△ **Translation T_2**
To move triangle $A_1B_1C_1$ to position $A_2B_2C_2$ each point moves 6 units horizontally, then moves 2 units vertically. The vector is written as above.

Direction of translations

The numbers used to show a translation's vector are positive or negative, depending on which direction the object moved. If it moves to the right or up, it is positive; to the left or down, it is negative.

▽ **Negative translation**
The rectangle ABCD, moves down and left, so the values in its vector are negative.



distance moved horizontally (to the left)

$$T_1 = \begin{pmatrix} -3 \\ -1 \end{pmatrix}$$

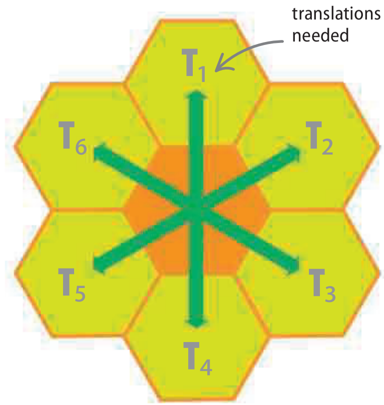
distance moved vertically (downward)

◁ **Translation T_1**
The translation T_1 moves rectangle ABCD to the new position $A_1B_1C_1D_1$. It is written as the vector shown—both its parts are negative.

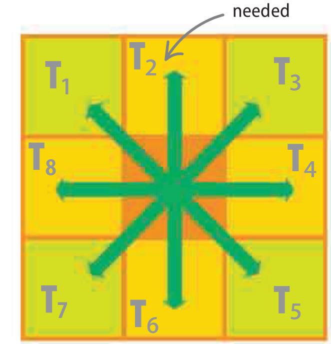
LOOKING CLOSER

Tessellations in action

A tessellation is a pattern created by using shapes to cover a surface without leaving any gaps. Two shapes can be tessellated with themselves using only translation (and no rotation)—the square and the regular hexagon. To tessellate a hexagon using translation requires 6 different translations; to tessellate a square requires 8.



△ **Hexagons**
Each of the hexagons around the outside is a translated image of the central hexagon. The tessellation continues in the same way.



△ **Squares**
Each of the squares around the edge is a translated image of the central square. The tessellation continues in the same way.